Databases and Web Programming Week 1 Lab

The objective of this lab is to develop a robust understanding of data structure by **deriving a schema from a given dataset**. You will learn to identify data types, constraints, and relationships, and document these findings in a formal schema.

What is a schema?

A **schema** is the blueprint of a database or dataset. It defines the structure of the data, including the names of all the columns (or fields), the data type of each column (e.g., integer, string, date), and any constraints that apply to the data (e.g., a column cannot be null, or a value must be unique). Creating a schema is a critical first step in data analysis, data warehousing, and database design. A well-defined schema ensures data integrity, consistency, and efficient querying.

Datasets

- Titanic Dataset
- Uber dataset

Procedure

1. **Inspect the Dataset**: Open the provided dataset. Begin by examining the first few rows to get an initial sense of the data. Look at the column headers and the values within each column. Pay attention to how the data is formatted.
2. **Identify Columns and Data Types**: For each column, determine the most appropriate data type. Consider the following:
    - **String**: Textual data, including names, descriptions, or codes.
    - **Integer**: Whole numbers (e.g., age, count).
    - **Float/Decimal**: Numbers with fractional values (e.g., price, temperature).
    - **Boolean**: True/False values.
    - **Date/Time**: Dates, times, or timestamps.
3. **Identify Constraints**: Look for rules or limitations on the data in each column.
    - **Not Null**: Are there columns that must always have a value?
    - **Unique**: Are there columns where every value must be different? (e.g., a username).
    - **Primary Key**: Is there a column that uniquely identifies each row? This is often an ID number.
    - **Foreign Key**: Is there a column that references a primary key in another table (if applicable)?

- **Domain/Value Constraints**: Are there specific rules about the values a column can hold? (e.g., a column for 'gender' might only contain 'M' or 'F').

4. **Document the Schema**: Record your findings in the table below. For each column, include the following information in a table or a clear list format:
    - **Column Name**: The name of the field.
    - **Data Type:** The determined data type (e.g., VARCHAR (50), INTEGER, DATE).
    - **Constraints:** Any constraints you identified (e.g., PRIMARY KEY, NOT NULL, UNIQUE).
    - **Description**: A brief, one-sentence description of the data contained in the column.
    - **Example Value**: A representative example from the dataset.

(Generate such table for each dataset)

Column name	Data type	Constrains	Description	Example value

Questions for Analysis

1. Why is identifying a **primary key** important for a schema?
2. Explain the difference between a **VARCHAR** and an **INTEGER** data type. When would you use one over the other?
3. How can a well-defined schema improve the performance of a database?
4. What are the potential consequences of having an inaccurate or incomplete schema?

Example of table schema documentation for task 4.

Driver Ratings	Float	Nullable, Range 0.0 - 5.0	The rating given to the driver by the customer.	4.9
Customer Rating	Float	Nullable, Range 0.0 - 5.0	The rating given to the customer by the driver.	4.9
Payment Method	Object	Nullable, Categorical	The payment method used for the ride.	"UPI"